

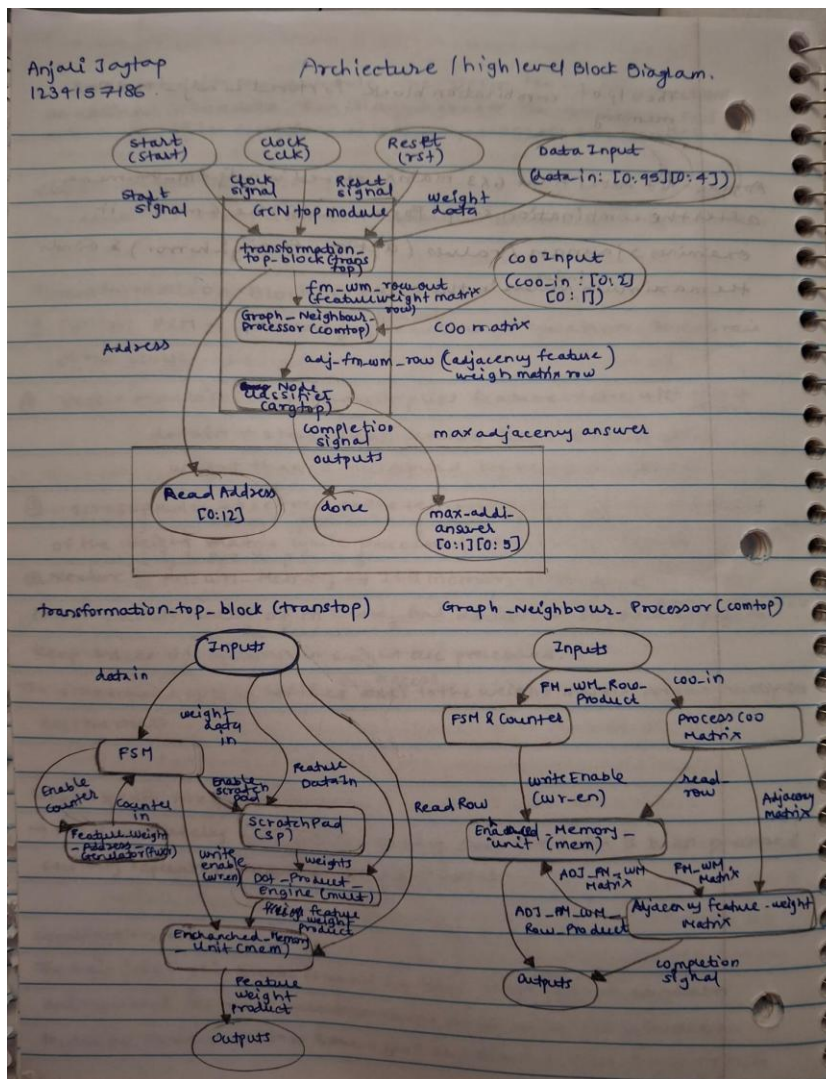
Name: Anjali Ajay Jagtap

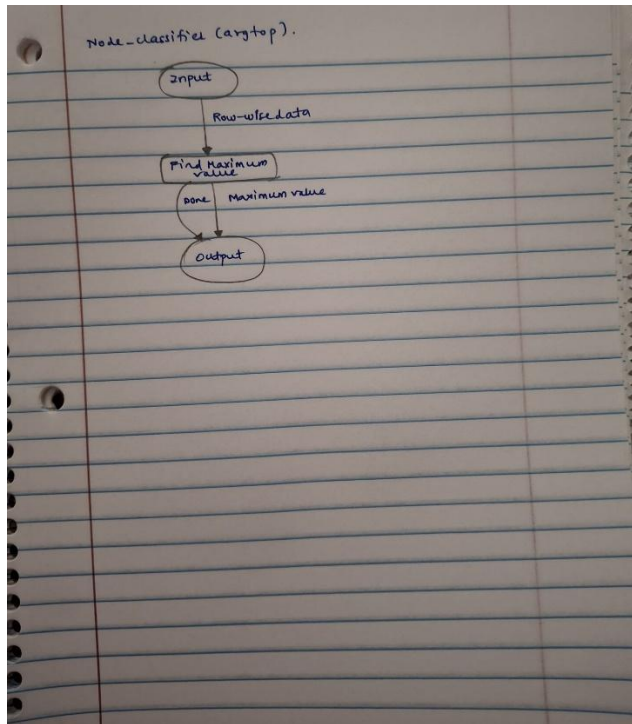
Asu id: 1234157186

EEE 525 Lab4

## Milestone 1

Architecture/High- Level Block Diagram :





## Design Decisions:

### 1. Modular Structure

- The architecture is organized into distinct SystemVerilog modules such as Transformation\_Block, Combination\_Block, and Argmax\_Block, each dedicated to a specific function.
- This modularization promotes code reusability, simplifies debugging, and enables independent verification of each module.

### 2. Efficient Memory Handling

- Intermediate data and computation outputs are stored in memory modules like Scratch\_Pad and Argmax\_Memory\_Logic.
- The design uses pipelined operations with careful planning and reset logic to avoid data hazards and ensure smooth, conflict-free execution.

### 3. Finite State Machine (FSM) Control

- FSMs like Transformation\_FSM and Argmax\_FSM govern the flow of computations, memory accesses, signal timing, address generation, and write-enable control.
- These FSMs guarantee that memory transactions occur only at the appropriate times, ensuring proper pipelined and sequential operation.

### 4. Argmax Computation Handling

- Specialized modules such as Argmax\_Comb\_Logic are responsible for performing

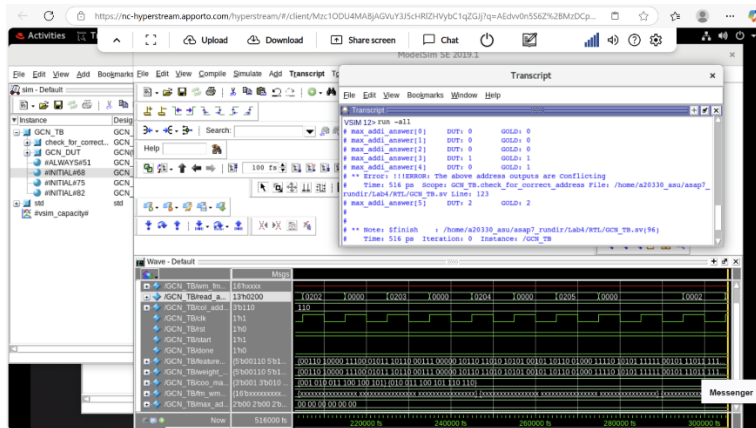
argmax operations, processing the results from matrix-vector multiplications efficiently.

**5. Multipliers & Adders used:**

I have done 12 element wise multiplication so  $12 * 8 \text{ segments} = 96$  multipliers used.

Adders:  $11 \text{ additions} * 8 \text{ segments} = 88 + 7 \text{ segments for final sum} \Rightarrow 88 + 7 = 95$   
adders used.

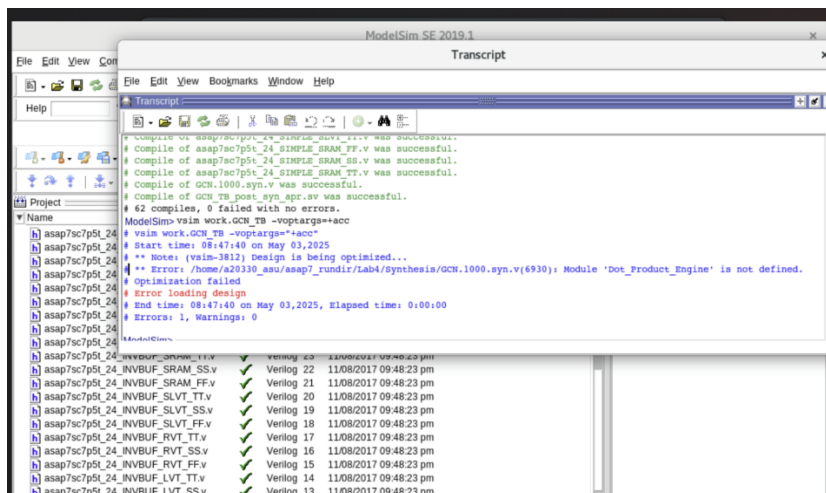
## Behavioral Verilog – Simulation



## Correctness of the module with behavioral Verilog

**Explanation to the error:** I have tried and checked the logic for argmax function like adding delays, latches , combinational and sequential logic ,tried checking reset logic but its reading correctly the index value for all other indexes except index its reading 0 instead of 1 its reading garbage value it's not updating after resetting even after adding delays.

### Post Synthesis -Simulation:



Correctness of the module with post-synthesis Verilog netlist

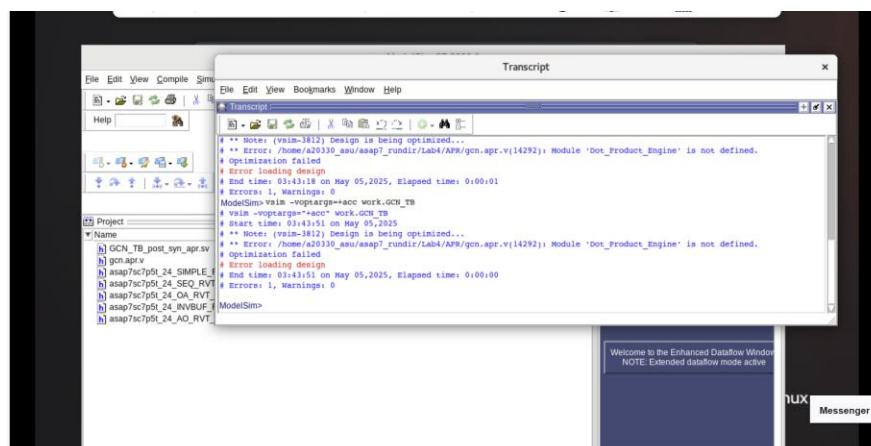
**Explanation of this error:** I have checked with TA he said your Dot Product Engine module is not synthesizing so it's not showing up in work directory unlike other modules when I run post synthesis, even after checking the logic blocking , non-blocking elements or combinational or sequential even after trying it's not synthesizing if that would have synthesize I would get exact same waveform like RTL

## Milestone 2

**Total Latency:**

o **Total Latency: 0.516 ns, at the clock frequency of 1000 MHz.**

o **Screenshot of Modelsim:**

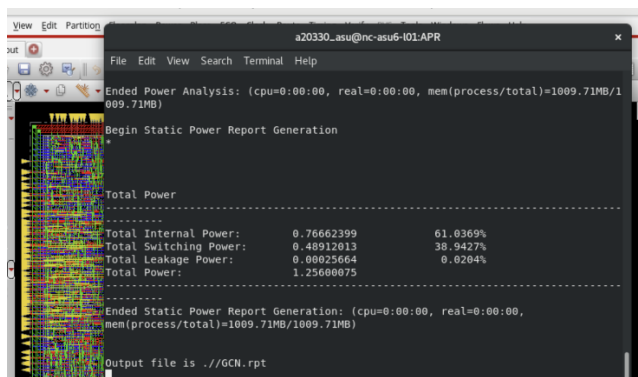


Post APR simulations screenshot

Explanation of this error: I have checked with TA he said your Dot Product Engine module is not synthesizing so it's not showing up in work directory unlike other modules when I run post synthesis, even after checking the logic blocking , non-blocking elements or combinational or sequential even after trying it's not synthesizing if that would have synthesize I would get exact same waveform like RTL for APR also same .

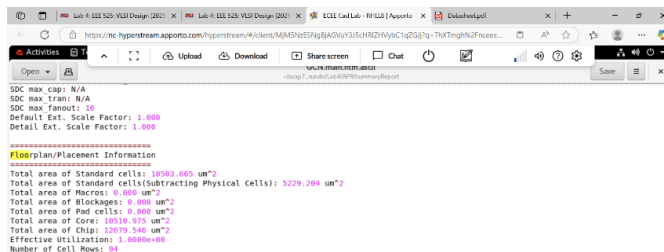
## Power:

- **Total Power(From Innovus): 1.2560mW**

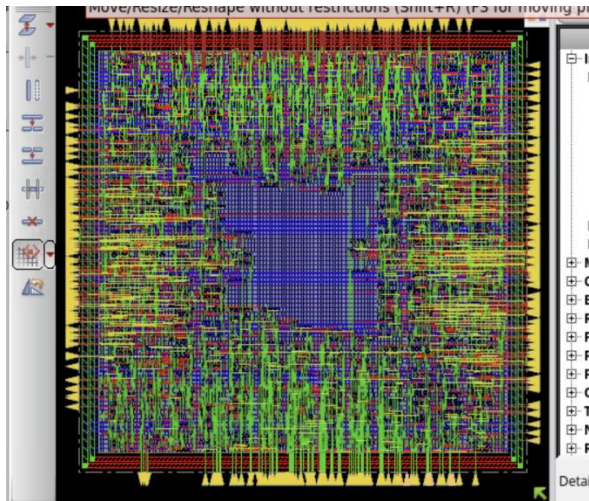


## Area:

- **Standard cells + Filler cells: 0.010503mm^2**



## Design x and y dimensions screenshot



## Innovus density:

o Before filler cell insertion: 53.22%

Linux X86\_64 (Host ID: nc-asub-118.apporto.com)  
 # Generated on: Wed Apr 30 03:45:41 2025  
 # Design: GCN  
 # Command: optDesign -postroute -hold

=====

optDesign Final SI Timing Summary

=====

| Hold Mode        | all   | regreg | default |
|------------------|-------|--------|---------|
| WNS (ns):        | 0.011 | 0.042  | 0.011   |
| TNS (ns):        | 0.000 | 0.000  | 0.000   |
| Violating Paths: | 0     | 0      | 0       |
| All Paths:       | 1043  | 511    | 1013    |

=====

| DRVs       | Real           |           | Total          |
|------------|----------------|-----------|----------------|
|            | Nr nets(terms) | Worst Vio | Nr nets(terms) |
| max_cap    | 0 (0)          | 0.000     | 1 (1)          |
| max_tran   | 0 (0)          | 0.000     | 0 (0)          |
| max_fanout | 95 (10)        | -56       | 96 (10)        |
| max_length | 0 (0)          | 0         | 0 (0)          |

Density: 53.22%  
 (Use 0% with Fillers)  
 Total number of glitch violations: 0

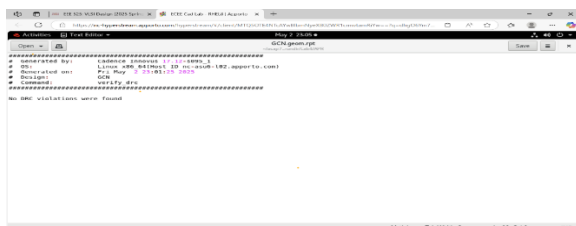
## Number of gates:

o Gates = 7472

o Cells = 2317

```
innovus 1> reportGateCount
Gate area 0.6998 um^2
[0] GCN Gates=7472 Cells=2317 Area=5229.2 um^2
innovus 2>
```

## Post\_APR – DRC Check:



No DRC violations

The screenshot shows a terminal window with the following content:

```

=====
# Generated by: Gcloud Instance ID: ml-asc-182.aypote.com
# OS: Linux x86_64 Instance ID: ml-asc-182.aypote.com
# Generated on: Fri May 3 10:45:18 UTC 2025
# Disks: GCE
# Command: verifyConnectivity
=====
Verify Connectivity Report is created on Fri May 3 10:45:18 UTC 2025

Begin Summary
Found no problems or warnings.
End Summary
  
```

### Worst Setup Path:

[illegible]

**Worst Setup, Timing tab**

| Path                             | Delay   | Setup   | Slack   |
|----------------------------------|---------|---------|---------|
| transtop/sp/FE_D0C8_scrtd_psd/A  | 8.000   | 270.000 | 496.221 |
| transtop/sp/FE_D0C8_scrtd_psd/V  | 87.700  | 358.500 | 583.921 |
| transtop/sp/FE_D0C11_scrtd_psd/A | 1.000   | 359.500 | 584.921 |
| transtop/sp/FE_D0C11_scrtd_psd/V | 91.200  | 450.000 | 676.021 |
| transtop/sp/FE_D0C14_scrtd_psd/A | 8.400   | 459.000 | 684.421 |
| transtop/sp/FE_D0C14_scrtd_psd/V | 128.000 | 587.000 | 812.421 |
| transtop/sp/FE_D0C15_scrtd_psd/A | 2.000   | 580.000 | 814.421 |
| transtop/sp/FE_D0C15_scrtd_psd/V | 91.600  | 680.000 | 906.021 |
| transtop/sp/u1412/A1             | 32.000  | 710.700 | 936.121 |
| transtop/sp/u1412/V              | 32.000  | 742.000 | 968.321 |
| transtop/sp/m617                 | 0.100   | 743.000 | 968.421 |
| transtop/sp/u1413/V              | 96.000  | 759.200 | 984.621 |
| transtop/sp/memory_reg8_2/_D     | 0.100   | 759.200 | 984.621 |

**Clock Rise Edge**

Beginpoint Arrival Time: 0.000

End Point Path: transtop/sp/memory\_reg8\_2/\_D

| Pin                            | Edge | Net | Cell                     | Delay | Arrival Time | Required Time |
|--------------------------------|------|-----|--------------------------|-------|--------------|---------------|
| clk                            | ^    | clk | ASYNC_DFFH11_ASA7P_75T_R | 0.000 | 0.000        | -225.421      |
| transtop/sp/memory_reg8_2/_CLK | ^    | clk | ASYNC_DFFH11_ASA7P_75T_R | 0.000 | 0.000        | -225.421      |

Path 2: NET Setup Check Pin 1522000/sp/memory\_reg8\_2/\_CLK

Endpoint: transtop/sp/memory\_reg8\_2/\_D (\*) checked with leading edge of 'clk'

Beginpoint: transtop/tfs/current\_state\_reg\_2/\_QN (\*) triggered by leading edge of 'clk'

Path Groups: [reg]reg

Analysis View: default setup view

Other End Arrival Time: 0.000

Setup: 0.000

Phase Shift: 1000.000

Required Time: 984.134

Arrival Time: 759.300

Slack Time: 225.834

Clock Rise Edge: 0.000

Beginpoint Arrival Time: 0.000

| File Edit View Settings              |   |                  |  |                          |  |      |  |              |         |
|--------------------------------------|---|------------------|--|--------------------------|--|------|--|--------------|---------|
| W                                    |   | R                |  | E                        |  | D    |  | S            |         |
| Pin                                  |   | Edge             |  | net                      |  | Cell |  | Delay        |         |
| Timing Path:                         |   |                  |  |                          |  |      |  | Arrival Time |         |
| clk                                  | ^ | clk              |  |                          |  |      |  | 0.000        | 226.287 |
| transpof/fsm/current_state_reg_2/CLK | ^ | clk              |  | ASYNC_DFFH11_A56P7_T5T_R |  |      |  | 0.000        | 226.287 |
| transpof/fsm/current_state_reg_2/Q0  | ^ | clk              |  | ASYNC_DFFH11_A56P7_T5T_R |  |      |  | 0.000        | 226.287 |
| transpof/fsm/ctrl_scdh_psd/a         | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/b         | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/c         | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/d         | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/e         | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/f         | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/g         | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/h         | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/i         | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/j         | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/k         | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/l         | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/m         | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/n         | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/o         | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/p         | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/q         | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/r         | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/s         | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/t         | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/u         | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/v         | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/w         | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/x         | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/y         | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/z         | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/aa        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/ab        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/ac        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/ad        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/ae        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/af        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/ag        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/ah        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/ai        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/aj        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/ak        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/al        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/am        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/an        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/ao        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/ap        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/aq        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/ar        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/as        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/at        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/au        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/av        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/aw        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/ax        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/ay        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/az        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/ba        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bb        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bc        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bd        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/be        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bf        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bg        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bh        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bi        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bj        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bk        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bl        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bm        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bn        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bo        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bp        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bq        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bd        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/be        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bf        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bg        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bh        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bi        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bj        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bk        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bl        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bm        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bn        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bo        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bp        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bq        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bd        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/be        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bf        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bg        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bh        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bi        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bj        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bk        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bl        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bm        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bn        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bo        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bp        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bq        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bd        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/be        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bf        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bg        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bh        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bi        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bj        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bk        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bl        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bm        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bn        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bo        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bp        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bq        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bd        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/be        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bf        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bg        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bh        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bi        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bj        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bk        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bl        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bm        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bn        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bo        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bp        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bq        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/bd        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T_R         |  |      |  | 0.000        | 64.400  |
| transpof/fsm/ctrl_scdh_psd/be        | ^ | transpof/fsm/n50 |  | NAND2_56P7_T5T           |  |      |  |              |         |

```

Wolfram Setup, Timing List
Wolfram Setup, Timing List
File Edit View
*****
# Generated by: Cadence Innovus 17.12-0091.1
# OS: Linux x86_64 (vst-10 uc-astud-110.apsports.com)
# Generated on: Wed Apr 30 03:40:41 2025
# Design: GCS
# Command: optDesign -postroute -hold
*****
Path 1: NET Removal Check with Pin transtop/memory_reg_42_r/CLK
Inputnet: transtop/memory_reg_42_r/SET (v) checked with leading edge of
CLK
Reginputset: reset (v) triggered by leading edge of
CLK
Path Groups: (async_default)
Analysis View: default_hold_view
Other End Arrival Time 0.000
+ Removal 5.124
+ Phase Shift 0.000
+ Required Time 5.124
+ Arrival Time 16.000
Sleak Time 11.476
Clock Rise Edge 0.000
+ Input Delay 0.100
+ Drive Adjustment 1.000
+ Reginputset Arrival Time 2.000
Timing Path:
*****
Pin Edge Net Cell Delay Arrival Required
Time Time Time
reset v reset BUFX2_ASAP7_T5T_R 0.000 2.000 5.476
FE_MHC3_reset/v v reset BUFX2_ASAP7_T5T_R 0.000 2.000 5.476
FE_MHC3_reset/v v FE_MHC3_reset BUFX2_ASAP7_T5T_R 14.000 16.000 5.124
FE_MHC3_reset/v v FE_MHC3_reset ADVNC_FSM1_ASAP7_T5T_R 0.000 16.000 5.124
*****
Clock Rise Edge 0.000
+ Reginputset Arrival Time 0.000
Other Path:
*****

```

[illegible]

|                              | Pin | Edge | met | Cell                    | Delay | Arrival Time | Required Time |
|------------------------------|-----|------|-----|-------------------------|-------|--------------|---------------|
| clk                          | clk |      |     |                         | 0.000 | 0.000        | 14.235        |
| transipio/memory_reg_1_1_clk | clk |      |     | ASYNC_SFHWIG_A6P7_7YS_R | 0.000 | 0.000        | 14.235        |



## Virtuoso LVS:

